Journal of Experimental Botany, Vol. 55, No. 407,
Water-Saving Agriculture Special Issue, pp. 2447–2460, November 2004
doi:10.1093/jxb/erh277 Advance Access publication 8 October, 2004

Journal of Experimental Botany

Breeding for high water-use efficiency

A. G. Condon[1,*], R. A. Richards[1], G. J. Rebetzke[1] and G. D. Farquhar[2]

[1] *CSIRO Plant Industry, GPO Box 1600, Canberra, ACT 2601, Australia*

[2] *Research School of Biological Sciences, Australian National University, Canberra, ACT 2601, Australia*

Received 3 June 2004; Accepted 26 August 2004

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

Abstract

There is a pressing need to improve the water-use efficiency of rain-fed and irrigated crop production. Breeding crop varieties with higher water-use efficiency is seen as providing part of the solution. Three key processes can be exploited in breeding for high water-use efficiency: (i) moving more of the available water through the crop rather than it being wasted as evaporation from the soil surface or drainage beyond the root zone or being left behind in the root zone at harvest; (ii) acquiring more carbon (biomass) in exchange for the water transpired by the crop, i.e. improving crop transpiration efficiency; (iii) partitioning more of the achieved biomass into the harvested product. The relative importance of any one of these processes will vary depending on how water availability varies during the crop cycle. However, these three processes are not independent. Targeting specific traits to improve one process may have detrimental effects on the other two, but there may also be positive interactions. Progress in breeding for improved water-use efficiency of rain-fed wheat is reviewed to illustrate the nature of some of these interactions and to highlight opportunities that may be exploited in other crops as well as potential pitfalls. For C_3 species, measuring carbon isotope discrimination provides a powerful means of improving water-use efficiency of leaf gas exchange, but experience has shown that improvements in leaf-level water-use efficiency may not always translate into higher crop water-use efficiency or yield. In fact, the reverse has frequently been observed. Reasons for this are explored in some detail. Crop simulation modelling can be used to assess the likely impact on water-use efficiency and yield of changing the expression of traits of interest. Results of such simulations indicate that greater progress may be achieved by pyramiding traits so that potential negative effects of individual traits are neutralized. DNA-based selection techniques may assist in such a strategy.

Key words: Carbon isotope discrimination, drought resistance, transpiration efficiency, wheat.

Introduction

Raising the water-use efficiency of both irrigated and rain-fed crop production is an urgent imperative (Hamdy *et al*., 2003). Of the world's allocatable water resource, *c*. 80% is currently consumed by irrigated agriculture. This level of consumption by agriculture is not sustainable into the future. Projected population growth (another 2 billion people within 2–3 decades) will require that more of the available water resource be used for domestic, municipal, industrial, and environmental needs. The most realistic solution to the increased demand for water will be reallocation to these other purposes of some of the water currently used by agriculture. Even a modest reallocation, reducing agriculture's share to 70%, would increase the amount of water available for other purposes by up to 50% (Hamdy *et al*., 2003). However, the expanded population will not only need more water to satisfy these other purposes, it will also need to be fed and clothed. This will require substantially more efficient production from a smaller irrigation water resource. It will also require substantially

* To whom correspondence should be addressed. Fax: +61 2 6246 5399. E-mail: tony.condon@csiro.au

Abbreviations: *A*, instantaneous rate of CO_2 assimilation; *A*/*T*, instantaneous water-use efficiency of leaf gas exchange; C_a, atmospheric CO_2 concentration; C_i, leaf internal CO_2 concentration; C_i/C_a, the ratio of internal and atmospheric CO_2 concentrations; $\Delta^{13}C$, carbon isotope discrimination; *E*, evaporation from the soil; *ET*, evapotranspiration, the sum of soil evaporation and plant transpiration from a crop canopy (instantaneous or cumulative); *G*, stomatal conductance; *HI*, harvest index; *T*, instantaneous rate of transpiration from a single leaf, and transpiration from a crop canopy (instantaneous or cumulative); *T*/*ET*, the ratio of transpiration to evapotranspiration; *W*, transpiration efficiency of crop biomass production; W_a, atmospheric water vapour concentration; W_i, leaf internal water vapour concentration.

 © Society for Experimental Biology 2004; all rights reserved

higher water-use efficiency from rain-fed agriculture, which remains the primary means of food production in most countries and for most farmers. Several strategies will be required to improve the productivity of water use in irrigated and rain-fed agriculture (Wang *et al*., 2002). Breeding crop varieties that are more efficient in their water use is one such strategy. Others include better management of the water resource and changes in crop management. None of these strategies should be seen as operating in isolation. Rather, it is likely that the greatest gains will be obtained through complementary approaches involving each of them.

In this article, recent progress in breeding for high water-use efficiency will be reviewed and some possible avenues for making further advances will be outlined. As a starting point, the article will first establish a conceptual framework for considering ways by which crop water-use efficiency might be improved through breeding. The prospects for improving crop water-use efficiency by changing the water-use efficiency of leaf gas exchange will be considered in some detail. Other likely avenues will also be considered briefly where substantial gains in crop water-use efficiency could be made. The review will concentrate on cereals and particularly on wheat, a dominant food crop which is grown in irrigated and rain-fed production systems over a wide range of latitudes. Wheat is also a crop that has been the focus of a long-standing breeding effort for higher water-use efficiency (Richards *et al*., 2002). This effort has culminated in the recent release in Australia of two new commercial wheat varieties, 'Drysdale' and 'Rees', bred specifically for higher water-use efficiency.

Water-use efficiency as a breeding target

Breeding to address a specific objective implies first, that the objective has been well defined and, second, that heritable traits have been identified that can come some way towards achieving the breeding objective. 'Water-use efficiency' as a breeding target could be defined in many ways, depending on the scale of measurement and the units of exchange being considered. All potential definitions will have some measure of water being exchanged for some unit of production. For physiologists, the basic unit of production could be moles of carbon gained in photosynthesis (A) in exchange for water used in transpiration (T). Thus a physiological definition might equate, at its most basic level, to the instantaneous water-use efficiency of leaf gas exchange (A/T). For farmers and agronomists, the unit of production is much more likely to be the yield of harvested product achieved from the water made available to the crop through precipitation and/or irrigation, i.e. a farmer's definition is one of agronomic water-use efficiency.

While agronomic water-use efficiency will be taken to be the ultimate breeding target, a major thrust of this article will be to place the physiological definition of water-use efficiency in the context of the farmer's definition. To do this it is useful to consider crop yield as being constructed from a framework of relatively simple components (equation 1).

$$\text{Yield}=ET\times T/ET\times W\times HI \quad (1)$$

In this framework, grain yield is described as being a function of the amount of water used by the crop (evapotranspiration, ET), the proportion of that water actually transpired by the crop (T/ET), the transpiration efficiency of biomass production (W), i.e. how much biomass is produced per millimetre of water transpired, and, lastly, how effectively the achieved biomass is partitioned into the harvested product, i.e. the ratio of grain yield to standing biomass termed the harvest index (HI). This framework is not based on the notion of 'drought resistance', but rather on the broad processes by which crops actually achieve yield in water-limited environments (Passioura, 1977; Condon and Richards, 1993; Richards *et al*., 2002). None of the components of this yield framework is truly independent of the others (Condon and Richards, 1993), but each can be considered a target for genetic improvement. Leaf-level water-use efficiency, A/T, is directly related to only one of these components, W, the transpiration efficiency of biomass production. However, as will be discussed in following sections, A/T also has the potential to influence each of the other three components in the yield framework.

Breeding for greater leaf-level water-use efficiency

The prospect of improving agronomic water-use efficiency by breeding for greater leaf-level water-use efficiency has long been an attractive one (Fischer, 1981). To assist in identifying ways this might be achieved, leaf-level water-use efficiency, A/T, can be described mathematically by noting that, first, A is the product of stomatal conductance to CO_2 (G_c) and the gradient in concentration of CO_2 between the outside (C_a) and inside (C_i) of the leaf (equation 2)

$$A=G_c(C_a-C_i) \quad (2)$$

and, second, that T is the product of stomatal conductance to water vapour (G_w) and the gradient in concentration of water vapour from the inside (W_i) to the outside (W_a) of the leaf (equation 3)

$$T=G_w(W_i-W_a) \quad (3)$$

For CO_2, the concentration is greater in the air outside the leaf, while the reverse is true for water vapour. The ratio A/T then becomes (equation 4)

$$A/T=[G_c(C_a-C_i)]/[G_w(W_i-W_a)] \quad (4)$$

which can be simplified even further (equation 5) by noting that the ratio of the diffusivities of CO_2 and water vapour in air has a value of c. 0.6. Thus,

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

$$A/T \approx 0.6C_a(1-C_i/C_a)/(W_i-W_a) \quad (5)$$

Equation (5) indicates two possible routes for improving leaf-level water-use efficiency. One is to lower the value of C_i/C_a, thereby increasing the value of $(1-C_i/C_a)$. The other is to make (W_i-W_a) smaller, i.e. to make the gradient driving transpirational water loss smaller.

Lowering the gradient in water vapour concentration

The simplest and most influential means by which breeding has improved the transpiration efficiency of biomass production via greater *A*/*T* has been to change crop characteristics so as to lower the average evaporative gradient during the crop growth cycle (Tanner and Sinclair, 1983; Richards *et al*., 2002). Reflecting processes at the leaf level, crop water loss is driven by the gradient in water vapour concentration between the crop canopy and the atmosphere. This gradient is least in cool, humid regions and, in most regions, during the coolest months of the year. During the past century breeders of many crop species have exploited genetic variation associated with intrinsic earliness, response to photoperiod, and vernalization requirement to generate enormous variation in crop phenology. This phenological variation has allowed crops to be grown successfully in regions and at times of the year that lower the prevailing evaporative demand, thus raising *A*/*T* and boosting crop yield. As opportunities arise, every effort should be taken to exploit this simple route to improved crop water-use efficiency further. Even seemingly unrelated objectives, such as eliminating disease susceptibility so that crops can be grown reliably in cooler, more humid conditions may present opportunities to adjust sowing time and crop phenology for improved *A*/*T* and crop yield. A good example of what is possible is the doubling of yield achieved by improving the disease resistance of chickpea, transforming it from a spring-sown to an autumn-sown crop in northern Syria (Keatinge and Cooper, 1983).

Changing the value of C_i/C_a

Referring back to equation (5), another way that breeding could improve *A*/*T*, and thereby improve the transpiration efficiency of biomass production, is to raise the value of the numerator $(1-C_i/C_a)$, i.e. to select genotypes for which the ratio C_i/C_a is small. A small value of C_i/C_a will reflect either a relatively low value of *G*, a relatively high photosynthetic capacity (amount and activity of photosynthetic machinery per unit leaf area) or a combination of these two (Farquhar *et al*., 1989). The interrelationships between C_i/C_a, photosynthetic capacity, and stomatal conductance are perhaps best appreciated in the context of the '*A*/C_i' plot (Long and Bernacchi, 2003). In such a plot (Fig. 1), curved lines rising from near the origin describe the dependence of *A* on C_i as external CO_2 concentration is varied experimentally. Variation in the initial slope of these curves equates to variation in photosynthetic capacity. In Fig. 1, the two curved lines

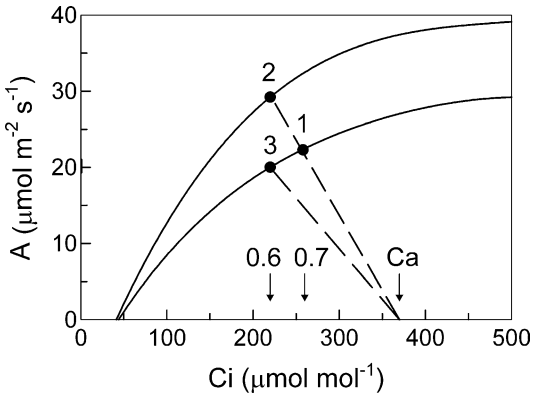


Fig. 1. Variation in the response of *A* to a change in C_i. The plotted curves are derived from leaf gas-exchange measurements on wheat genotypes reported in Condon *et al*. (1990). Data points at ambient C_a are plotted for three 'genotypes'. These data points indicate that lower values of C_i (and of C_i/C_a), and thus greater values of *A*/*T*, may be achieved through either greater photosynthetic capacity (indicated by a steeper initial slope of the solid lines originating near the origin) or by lower stomatal conductance (indicated by a smaller slope in the dashed lines originating at C_a). Changes in C_i/C_a can also be achieved through changes in both capacity and conductance (adapted from Condon and Hall, 1997. Reprinted with permission from Elsevier.).

are intersected by two straight lines originating at the ambient CO_2 concentration, C_a. Variation in the slope of these straight lines equates to variation in stomatal conductance. The intersections indicated by numerals represent the operating values of C_i (and thus C_i/C_a, *A*, and *G*) for three genotypes of a C_3 species. As shown in Fig. 1, lower operating values of C_i/C_a (say 0.6 compared with 0.7) may be achieved through higher photosynthetic capacity (genotype '2' compared with genotype '1'), lower stomatal conductance (genotype '3' compared with genotype '1'), or a combination of these two.

It is clear from Fig. 1 that it is not possible to infer, merely from measuring a change in C_i/C_a, that such a change is due to a change in photosynthetic capacity or to a change in *G*. Two other things are also clear from Fig. 1. If C_i/C_a is lowered as a result of an increase in photosynthetic capacity, then *A*, the rate of photosynthesis per unit leaf area, will be greater. Conversely, if C_i/C_a is lowered as a result of a decrease in *G*, then the value of *A* will be lowered.

There is likely to be another penalty, not revealed in Fig. 1, associated with a decrease in *G*. Data for '*A*/C_i' plots, and in fact most leaf gas-exchange data, are collected by inserting leaves into cuvettes in which the air is very well stirred and the leaf-air vapour concentration gradient is constrained to be approximately constant. Outside such cuvettes, this is unlikely to be the case, with the result that as *G* decreases leaf temperature will increase. An increase in leaf temperature will cause an increase in W_i and therefore an increase in the gradient driving transpiration (W_i-W_a), resulting in an increase in *T* per unit change in *G*. The net result is that the change in *A*/*T* as *G* decreases will not be as great as that predicted by equation (5).

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

The relationship between carbon isotope discrimination and C_i/C_a

Despite the advent of reliable, relatively-portable leaf gas-exchange systems, it is pertinent to note that *G*, photosynthetic capacity, and *A*/*T* are still tedious to measure in large breeding populations. It is now accepted that relative differences in C_i/C_a, at least within C_3 species, may be evaluated indirectly by measuring the carbon isotope composition of plant dry matter (Farquhar *et al*., 1989). The stable isotope of carbon, ^{13}C, makes up very close to 1% of the carbon in atmospheric CO_2. The proportion of ^{13}C in the dry matter of C_3 plants is fractionally less than in the atmosphere, primarily because C_3 species discriminate against ^{13}C during photosynthesis. Carbon isotope discrimination ($\Delta^{13}C$) is a measure of the $^{13}C/^{12}C$ ratio in plant material relative to the value of the same ratio in the air on which plants feed, and has been defined as follows (Farquhar and Richards, 1984):

$$\Delta^{13}C=[(R_a/R_p)-1]\times 1000 \quad (6)$$

where R_a is the value of the $^{13}C/^{12}C$ ratio in the atmosphere and R_p is the value of $^{13}C/^{12}C$ in plant material. For convenience, units of $\Delta^{13}C$ are expressed as per thousand (‰), i.e. the fractional difference from unity is multiplied by 1000. The ratio R_a/R_p has a value near 1.02 for C_3 plants, giving values of $\Delta^{13}C$ near 20‰.

There are several processes that contribute to the value of $\Delta^{13}C$ measured in plant dry matter of C_3 species (Farquhar *et al*., 1989; Brugnoli and Farquhar, 2000). The following approximate expression (equation 7) was developed by Farquhar and Richards (1984) to account for the two dominant processes; fractionations associated with diffusion of CO_2 into the intercellular airspaces (a=4.4‰) and fractionations associated with carboxylation of CO_2 into the first products of photosynthesis by Rubisco (b≈28‰).

$$\Delta^{13}C=a+(b-a)C_i/C_a \quad (7)$$

The value of 28‰ for b is approximate because there are several additional potential sources of variation in $\Delta^{13}C$, such as those associated with subsequent metabolism and respiration, but these are ill-defined and in most situations are likely to be small in magnitude (Brugnoli and Farquhar, 2000). Substituting the numerical values indicated into equation (7) generates the following:

$$\Delta^{13}C\approx 4.4+23.6C_i/C_a \quad (8)$$

This simple equation shows that $\Delta^{13}C$ is positively related to the ratio C_i/C_a. However, earlier it was noted that *A*/*T* should be negatively related to C_i/C_a (equation 5). Therefore $\Delta^{13}C$ and *A*/*T* should also be negatively related.

Carbon isotope discrimination and water-use efficiency

The realization that $\Delta^{13}C$ could provide a relatively simple, indirect measure of variation in *A*/*T* (Farquhar *et al*., 1982; Farquhar and Richards, 1984) gave renewed impetus to the prospect of exploiting variation in leaf-level water-use efficiency to improve agronomic water-use efficiency. Up to that time, it had been considered that there was little variation in C_i/C_a within or among C_3 crop species, the only substantial difference being that between C_3 and C_4 species (Fischer, 1981; Tanner and Sinclair, 1983). Since the pioneering work of Farquhar and colleagues (Farquhar *et al*., 1982; Farquhar and Richards, 1984), it has subsequently been demonstrated, for several C_3 species, that variation in $\Delta^{13}C$ closely reflects variation in the ratio C_i/C_a. One such example is shown in Fig. 2A, but there have been many similar studies (reviewed in Brugnolli and Farquhar, 2000) in which either $\Delta^{13}C$ was measured 'on-line' by sampling the air stream before and after it passed over a leaf, $\Delta^{13}C$ was measured in recent photosynthate extracted from leaves, or $\Delta^{13}C$ was measured in the dry matter of sink tissues.

Variation in $\Delta^{13}C$ among genotypes of C_3 species is large enough to, in theory, generate substantial variation in

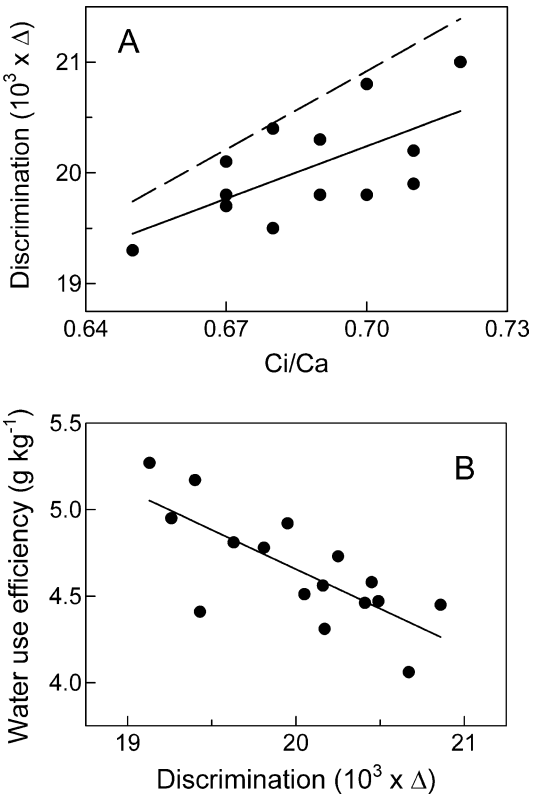


Fig. 2. (A). Relationship between C_i/C_a measured on flag leaves and $\Delta^{13}C$ measured in dry matter of growing ears of wheat enclosed in the flag leaf sheath, for 13 genotypes. The growing ear is a major sink for C assimilated by the flag leaf. The solid line is a linear regression fitted to the data points. The dashed line is the relationship $\Delta^{13}C=4.4+23.6C_i/C_a$ (equation 8) (adapted from Condon *et al*., 1990 with permission from CSIRO Publishing.). (B). Relationship between plant water-use efficiency (g dry matter produced kg^{-1} water transpired) and $\Delta^{13}C$ of plant dry matter, for 16 wheat varieties grown without water limitation in large pots sealed at the soil surface to eliminate soil evaporation. The solid line is a linear regression fitted to the data points (adapted from Condon *et al*., 1990 with permission from CSIRO Publishing.).

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

A/T and potentially, substantial variation in water-use efficiency of dry matter production (Farquhar and Richards, 1984). This has been confirmed in numerous studies with pot-grown plants. Negative correlations between $\Delta^{13}C$ and plant water-use efficiency have been demonstrated in many species (Fig. 2B), as predicted by theory (reviewed in Farquhar *et al*., 1989; Hall *et al*., 1994; Condon and Hall, 1997). Further, it has been shown that with adequate attention to sampling strategies, $\Delta^{13}C$ is a highly heritable trait that is relatively easy to manipulate in breeding populations (Condon and Richards, 1992; Rebetzke *et al*., 2002). Measurements of $\Delta^{13}C$ can be taken using freshly-sampled or stored plant material and they can be automated for relatively rapid throughput. All these observations point to $\Delta^{13}C$ as a potential candidate for use in breeding for greater agronomic water-use efficiency via greater *A/T*. However, significant challenges have arisen as attempts have been made to 'scale up' from associations between $\Delta^{13}C$ and water-use efficiency of leaves and single plants to associations between $\Delta^{13}C$ and water-use efficiency and yield of field stands. Some of these challenges were anticipated, others were not. The nature of these challenges and possible solutions are considered in the following sections.

Relationships between grain yield and carbon isotope discrimination in wheat and barley

The greatest challenge to using $\Delta^{13}C$ in breeding for greater agronomic water-use efficiency is the high level of inconsistency observed in the relationship between $\Delta^{13}C$ and yield. This inconsistency has been well-documented in numerous studies involving the cereals bread wheat (*Triticum aestivum* L.), durum wheat (*T. turgidum* var. *durum* L.) and barley (*Hordeum vulgare* L.). From the negative association between $\Delta^{13}C$ and leaf-level *A/T* and the consistently negative associations observed between $\Delta^{13}C$ and water-use efficiency at the single plant level in many pot studies, it might be inferred that crop yield and $\Delta^{13}C$ might also be consistently negatively related. Yet for a large number of studies involving collections of cereal genotypes grown in rain-fed and irrigated environments in Australia (Condon *et al*., 1987, 1993, 2002; Condon and Richards, 1993; Condon and Hall, 1997), the Mediterranean region (Voltas *et al*., 1999; Merah *et al*., 2001; Royo *et al*., 2002; Araus *et al*., 2003) and elsewhere (Sayre *et al*., 1995; Fischer *et al*., 1998), relationships reported between grain yield and $\Delta^{13}C$ have only infrequently been negative. Much more often these relationships have been either positive or 'neutral'. Many of the studies on associations between productivity and $\Delta^{13}C$ in cereals have used sets of genotypes in which there has been substantial variation not only in $\Delta^{13}C$, but in flowering date and height as well, two characteristics that could strongly influence yield independent of variation in $\Delta^{13}C$. Even in studies where flowering date and height have been tightly constrained, such as that shown in Fig. 3 (discussed below), associations between productivity and $\Delta^{13}C$ have still been variable.

Low-$\Delta^{13}C$ cereals tend to have 'conservative' crop growth

Why have relationships between grain yield and $\Delta^{13}C$ been so variable in so many studies? There appear to be several reasons, but a critical one is that for cereals low-$\Delta^{13}C$ (high *A/T*) is a 'conservative' trait in terms of water use and, perhaps more importantly, in terms of crop growth rate. Put simply, in the absence of soil water deficit, low-$\Delta^{13}C$ genotypes tend to grow slower than high-$\Delta^{13}C$ genotypes, resulting in lower total biomass production and grain yield (Condon and Richards, 1993; Condon *et al*., 1993, 2002; Lopez-Casteneda *et al*., 1995). The data in Fig. 3 for isomorphic recombinant inbred lines grown in the favourable 1993 season illustrate this well. In this instance, positive associations of $\Delta^{13}C$ with biomass production and grain yield were sustained by substantial regular rainfall events throughout the growing season.

An obvious, widely-cited reason why low-$\Delta^{13}C$ could be associated with 'conservative' crop growth rate in cereals is

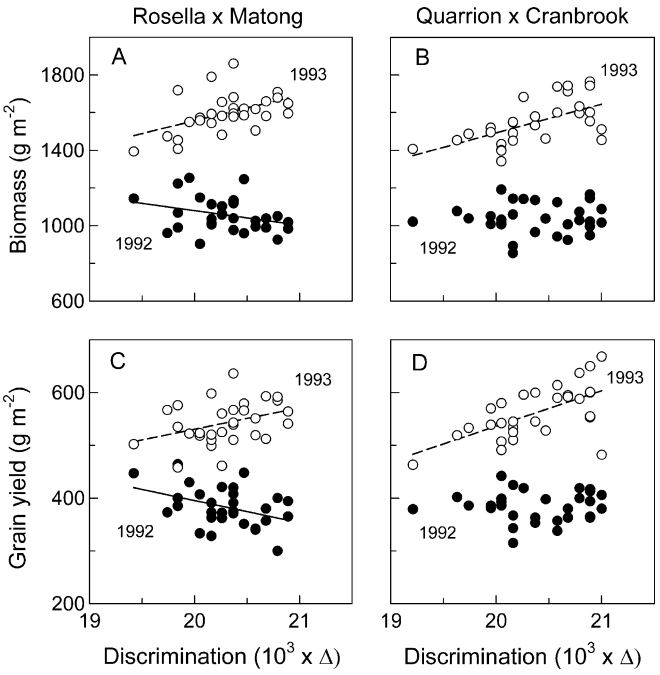


Fig. 3. Relationships between measures of crop productivity and $\Delta^{13}C$ for recombinant inbred lines of bread wheat grown under rainfed conditions at Condobolin, eastern Australia, in 1992 and 1993. Closed symbols are for 1992 and open symbols are for 1993. The lines were F_6 (1992) and F_7 (1993) progeny of crosses between parents with low and high values of $\Delta^{13}C$. The values of $\Delta^{13}C$ plotted are genotype means for leaf material of F_5 lines sampled early in the 1991 season when there was no soil water deficit. Least-squares linear regressions are shown for statistically-significant relationships (P <0.05, n=30) (adapted from Condon and Hall, 1997. Reprinted with permission from Elsevier.).

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

if differences in $\Delta^{13}C$ in the absence of soil water deficit are the result of differences in stomatal conductance. Genotypes with lower stomatal conductance will tend to have higher *A/T* and lower $\Delta^{13}C$, all else being equal, but in this case higher *A/T* is likely to be associated with lower photosynthetic rate per unit leaf area (Fig. 1, 'genotype' 3) and, consequently, a slower rate of crop growth.

Low conductance may not be the only reason why low-$\Delta^{13}C$ is associated with 'conservative' crop growth rate. In cereals, variation in $\Delta^{13}C$ can also result from variation in photosynthetic capacity (Condon *et al*., 1990; Morgan and LeCain, 1991). If low-$\Delta^{13}C$ is the result of high photosynthetic capacity, the expectation might be a higher rate of photosynthesis per unit leaf area (Fig. 1, 'genotype' 2) and thus faster crop growth rate. However, crop growth rate may again be slower because, in cereals, substantial increases in photosynthetic capacity are most readily achieved by concentrating N into smaller leaves that intercept less light. This may slow the rate of crop growth until full light interception is achieved (Condon and Richards, 1993; Condon and Hall, 1997). If full light interception is not achieved or achieved only briefly, as often occurs in drier cropping environments, then high photosynthetic capacity may not result in greater growth. The data in Fig. 4 indicate that, in fact, the reverse may occur, i.e. that cereal genotypes with low photosynthetic capacity may actually achieve faster crop growth. In this study, biomass production to anthesis was positively correlated with $\Delta^{13}C$ (Fig. 4A), as has been frequently observed, but the three genotypes with the highest $\Delta^{13}C$ values did not have greater *G* (Fig. 4B). Instead, the inference, from similar values of *G* but higher values of $\Delta^{13}C$, was that they had lower photosynthetic capacity. This inference was confirmed in studies on glasshouse-grown plants (Condon *et al*., 1990).

So, in cereals, higher photosynthetic capacity may not necessarily be associated with faster crop growth rate. A similar conclusion can be drawn from the study by Fischer *et al*. (1998) on physiological changes in bread wheats released by CIMMYT from 1962 to 1988. The more recent of these wheats had substantially higher grain yields under irrigation, but this was not associated with greater biomass production. The yield gain, however, reflected higher harvest index for the more recent wheats. Among this 'historic' collection of CIMMYT wheats there was no change in total biomass production despite the more recent wheats having both higher photosynthetic capacity and higher *G*, together generating substantially higher values of *A* on a leaf area basis.

Interactions between growth and water use

Irrespective of its physiological basis, 'conservative' crop growth by low-$\Delta^{13}C$ cereal genotypes has important implications for agronomic water-use efficiency. The most consistently positive relationships between $\Delta^{13}C$ and yield

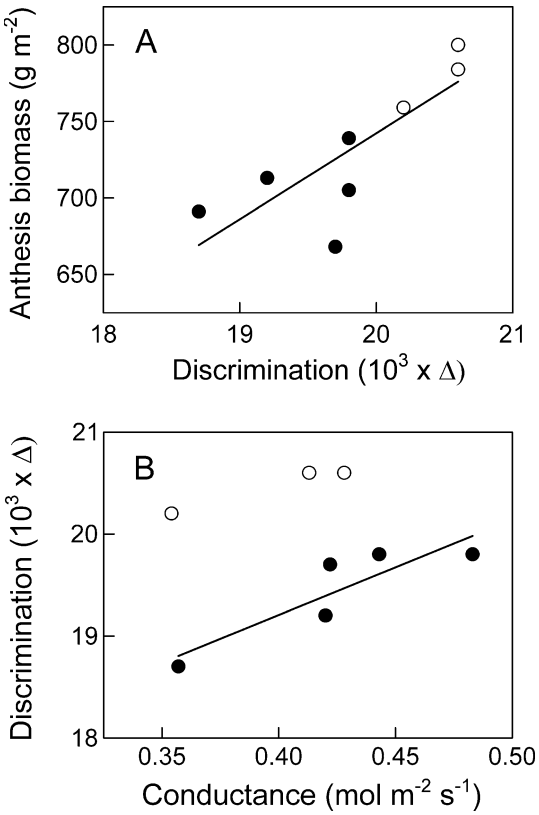


Fig. 4. (A) The relationship between above-ground biomass production to anthesis and leaf $\Delta^{13}C$, and (B) the relationship between leaf $\Delta^{13}C$ and stomatal conductance, for eight wheat genotypes grown at Moombooldool, eastern Australia in 1985. Values of $\Delta^{13}C$ are average values for leaves from well-watered plants sampled well before anthesis. Values of stomatal conductance are the average of data obtained at five sampling times before anthesis. Least-squares linear regressions are shown for statistically-significant relationships (P <0.05). Open symbols indicate genotypes with low photosynthetic capacity (adapted from Condon *et al*., 1990, 1993 with permission from CSIRO Publishing.).

have been found in environments or seasons where supplemental irrigation or regular rainfall events throughout the growing season have maintained a high soil water status. In these environments the faster growth of high-$\Delta^{13}C$ genotypes has usually translated directly into higher final biomass production and grain yield (Condon *et al*., 1987, 2002; Sayre *et al*., 1995; Fischer *et al*., 1998). Low-$\Delta^{13}C$ genotypes have achieved less biomass and lower yields. It is also likely that they have also left more water behind in the soil profile at maturity (Condon *et al*., 2002).

In less favourable environments, variation in the extent and timing of any water limitation may interact with the 'conservative' growth and water use of low-$\Delta^{13}C$ genotypes to generate complex relationships between yield and $\Delta^{13}C$. This complexity is also illustrated in Fig. 3. In the drier 1992 season, high-$\Delta^{13}C$ was associated with lower biomass production and grain yield in the lines from one cross (Rosella×Matong), but for lines from the second cross (Quarrion×Cranbrook) there were no associations

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

between productivity and $\Delta^{13}C$. Soil water status at flowering is likely to have been higher for the second cross because lines from this cross flowered, on average, one week earlier than lines from the first cross. The results for these two sets of lines grown in these two seasons indicate that the amount of rainfall is an important variable contributing to variation in grain yield. They also indicate that the timing of development of soil water deficit, with respect to the critical flowering phase, is also an important variable.

If high-$\Delta^{13}C$ genotypes exhaust the available soil water too quickly, before flowering, there is likely to be a yield penalty (Fig. 3C; Fischer, 1981). However, the penalty, in terms of crop water use and yield, associated with faster growth of high-$\Delta^{13}C$ cereals is not always as great as might be expected. In fact, the faster growth of high-$\Delta^{13}C$ cereal genotypes has often been shown to be of benefit in seasons or environments in which frequency of rainfall events was high early in the season, but was not sustained during later stages of growth, typical Mediterranean-type environments (Condon *et al*., 1993; Condon and Hall, 1997; Voltas *et al*., 1999; Merah *et al*., 2001; Royo *et al*., 2002; Araus *et al*., 2003). In this sort of environment, evaporation from the soil surface can account for as much as 50% of the growing season rainfall (Cooper *et al*., 1987). Studies where water use has been carefully partitioned between plants and soil have shown that the more 'profligate' transpiration associated with high-$\Delta^{13}C$ actually resulted in little difference in total water use to anthesis, despite substantially more growth by high-$\Delta^{13}C$ genotypes at this critical stage of development (Condon *et al*., 1993, 2002; Lopez-Castaneda *et al*., 1995). This was because the fast early growth of high-$\Delta^{13}C$ genotypes meant that they shaded the soil surface quickly. They were thus much more successful at restricting soil evaporation when rain events were frequent than were slower-growing, low-$\Delta^{13}C$ genotypes. Because total water use to anthesis was similar, the greater anthesis biomass of high-$\Delta^{13}C$ genotypes translated into higher grain number and higher yield.

In other environments high-$\Delta^{13}C$ genotypes have achieved greater anthesis biomass at the expense of substantially greater soil water depletion at anthesis (Condon *et al*., 1993). Yet despite achieving very little growth after flowering compared with low-$\Delta^{13}C$ genotypes that had been more conservative in their water use, the high-$\Delta^{13}C$ genotypes were still able to yield more. This was because high-$\Delta^{13}C$ genotypes achieved a larger grain number (associated with higher anthesis biomass) and were then able to fill these grain, probably by translocating large amounts of stored assimilate. In the same environments low-$\Delta^{13}C$ genotypes failed to use stored assimilates as effectively. They may have had fewer reserves, due to less anthesis biomass, or the stored assimilates may not have been necessary because the low-$\Delta^{13}C$ genotypes had subsoil moisture in reserve to sustain higher photosynthesis during grain filling (Condon and Hall, 1997). One outcome of achieving high yield, despite relatively little post-anthesis growth, is that high-$\Delta^{13}C$ genotypes tend to have a higher harvest index (*HI*).

Finally, there have been a small number of studies conducted in environments where there was relatively little rainfall during the growing season and crops relied heavily on soil moisture reserves from substantial rains before sowing or very early in crop development (Condon and Richards, 1993; Condon and Hall, 1997; Condon *et al*., 2002). It has been in these few studies, where there was relatively little soil evaporation and a strong reliance on metering-out soil-water reserves before anthesis, that negative associations between yield and $\Delta^{13}C$ have been most consistently observed.

To summarize these complex associations between $\Delta^{13}C$, growth, water use, and yield in cereals, the 'conservative' growth associated with low-$\Delta^{13}C$ and high *A*/*T* has been found to interact with the extent and timing of water limitation to generate confounding effects on all of the components of the yield framework introduced earlier (equation 1). In some studies, high crop *W* (reflecting high leaf-level *A*/*T*) has been offset by less total crop water use. High-$\Delta^{13}C$ genotypes yielded more and dried the soil profile more fully. Low-$\Delta^{13}C$ genotypes have also been found to lose more water by evaporation from the soil surface. In combination, these two factors of less soil-water extraction and greater losses to evaporation have meant that low-$\Delta^{13}C$ genotypes have sometimes been found to transpire considerably less water than high-$\Delta^{13}C$ genotypes (Condon *et al*., 1993, 2002). Thus there has often been no biomass gain associated with high *A*/*T* and, frequently, less biomass production and lower grain yield. Greater biomass production by high-$\Delta^{13}C$ genotypes has been most marked at anthesis, this difference at anthesis being subsequently reflected in higher grain number and yield of high-$\Delta^{13}C$ genotypes even when they may have shown little biomass growth after anthesis. With respect to grain yield, the positive associations between biomass production and $\Delta^{13}C$ have been amplified in some studies by positive associations between *HI* and $\Delta^{13}C$, perhaps partly due to greater translocation of stored reserves by high-$\Delta^{13}C$ genotypes.

Defining the target environment using simulation modelling

The confounding effects arising from the conservative growth of low-$\Delta^{13}C$ genotypes indicate that the potential yield benefit from breeding cereals with high *A*/*T* may be limited, and is certainly likely to vary substantially with growing environment (Condon and Richards, 1993; Richards and Condon, 1993; Condon *et al*., 2002; Araus *et al*., 2003). This conclusion was tested more closely by examining the potential outcome of breeding for high *A*/*T* using a wheat crop growth simulation model (Condon *et al*., 2002). The 'SIMTAG' model (Stapper and Harris, 1979)

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

was run for three representative, low-rainfall cropping environments in eastern Australia which were similar in average annual rainfall receipts (450–600 mm), but which varied in average seasonal distribution of rainfall, from winter-dominant to summer-dominant. In the first instance, for each location, the growth and yield of a 'standard', autumn-sown genotype was simulated using several decades of historic weather data. The model was then run again, for each location, to simulate the likely long-term average effect on yield of growing a genotype incorporating improved *A/T*, a genotype with faster early leaf area growth (to restrict soil evaporation), and a genotype combining both these traits. The principal results of the simulations were as follows (Fig. 5): (i) Incorporating higher *A/T* resulted in an average yield advantage of 11% in the northern, summer-dominant rainfall environment where the crop relies strongly on metering out stored subsoil moisture. There was almost no increase in average yield in the southern, winter-dominant rainfall environment nor in the central region with highly variable rainfall. (ii) Faster early leaf area growth was most important in the winter-dominant, Mediterranean-type rainfall environment in southern Australia. It had little benefit in the northern region, where soil evaporation is a relatively small component of total crop water use. Long-term average yield gains ranged from 11% in the south to only 1% in the north. (iii) Combining high *A/T* with faster early leaf area growth gave consistently large average yield gains across all three environments. Average yield gains ranged linearly from *c*. 17% in the south to 14% in the north. Combining the two traits provided the greatest advantage in the central region. In this region the rainfall distribution is uniform on average, but highly variable in amount and distribution from year to year.

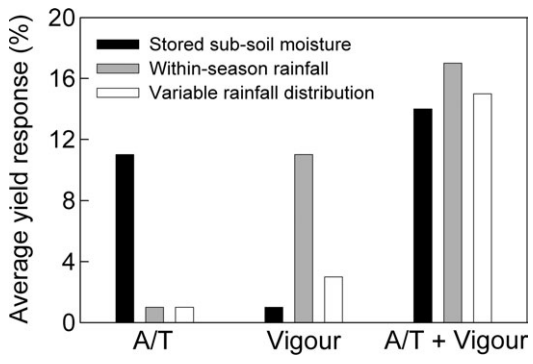


Fig. 5. Summary of results of wheat crop growth simulations using the 'SIMTAG' crop growth model and long-term weather data (30–50 years) showing the average change in yield, in three environments in eastern Australia, of breeding for either higher *A/T*, greater early vigour, or a combination of higher *A/T* and greater early vigour (Condon and Stapper, 1995; AG Condon *et al*., unpublished results). Simulations were done for autumn-sown crops grown in (i) a northern, sub-tropical environment where crops have a strong reliance on sub-soil moisture stored from summer rainfall, (ii) a southern environment with a winter-dominant rainfall pattern where the crop relies heavily on within-season rainfall, and (iii) an environment between these two, where the average rainfall distribution is even but rainfall is highly variable in amount and seasonal distribution.

Given the results of the many field experiments summarized in the previous section, the first two outcomes of this simulation study were not particularly surprising. They indicated that breeding for higher *A/T* is probably a reasonable objective for the northern, summer-dominant rainfall environment and that breeding for greater early vigour is probably a reasonable objective for the southern, winter-dominant rainfall region. The large yield responses in all regions from combining the two traits was more of a surprise. This outcome probably reflects the fact that even though there are large differences between regions in the long-term average distribution of rainfall, there is substantial year-to-year variation in rainfall total and rainfall distribution in all three regions. Is it possible to combine these two traits by breeding? In the previous section it was emphasized that, for cereals, low $\Delta^{13}C$ (high *A/T*) is typically associated with relatively slow growth. So it may be difficult, physiologically, to combine high *A/T* and fast early leaf area growth. But there may be ways to boost early growth that are independent of changes in $\Delta^{13}C$. These will be discussed in a later section.

Yield response to breeding for high A/T

Experience over many growing seasons at a range of locations in eastern Australia, supported by the outcomes of the simulation study, indicated that breeding for higher *A/T* could have a large average benefit in the northern cropping region of eastern Australia. In this region the wheat crop relies on moisture stored from summer rains and *T* makes up a large proportion of total crop *ET*. A backcrossing programme was initiated to improve *A/T* (i.e. lower the $\Delta^{13}C$) of the relatively high-$\Delta^{13}C$ variety Hartog, widely-grown in Australia's sub-tropical northern region. Briefly, Hartog was crossed with a low-$\Delta^{13}C$ donor, the lowest-$\Delta^{13}C$ F_3 families were selected and these were back-crossed with Hartog twice more, without selection between these two rounds of crossing.

The effect on yield of divergent selection at the BC_2 stage, based on measurements of $\Delta^{13}C$ was tested by growing 60 BC_2 lines with very similar height and flowering time in nine rain-fed environments in eastern Australia and five in Western Australia. Thirty of the BC_2 lines had high *A/T* (low $\Delta^{13}C$) and the other 30 had low *A/T* (high $\Delta^{13}C$). The results are summarized in Fig. 6A, which shows the relative average yield advantage of the low-$\Delta^{13}C$ selections at each location. The environments sampled in eastern Australia varied widely in the amount and seasonal distribution of rainfall and environment-mean yields ranged from 1.3 to 6.3 t ha^{-1}. High-*A/T* lines out-yielded low-*A/T* lines in all of these environments (Fig. 6; Rebetzke *et al*., 2002). The relative yield advantage of high-*A/T* lines over low-*A/T* lines increased linearly as site-mean yield declined. The relative yield advantage was *c*. 10% in the

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

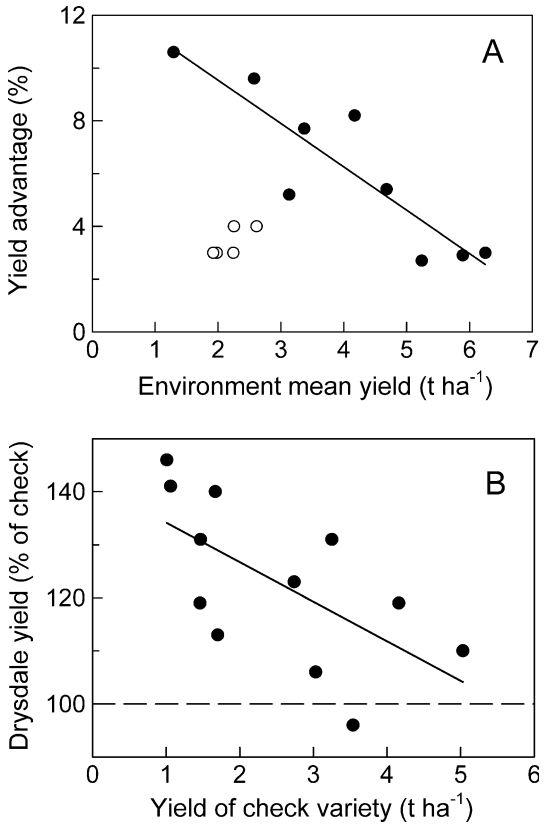


Fig. 6. (A) Average grain yield advantage of BC_2 bread wheat breeding lines selected for low-$\Delta^{13}C$ compared with BC_2 breeding lines selected for high-$\Delta^{13}C$, grown in nine environments in eastern Australia (closed symbols) and five environments in Western Australia (open symbols) from 1995 to 1998. Grain yield advantage of low-$\Delta^{13}C$ breeding lines was calculated as [(average yield of low-$\Delta^{13}C$ selections)–(average yield of high-$\Delta^{13}C$ selections)]/[average yield of high- $\Delta^{13}C$ selections] (adapted from Condon *et al*., 2002 with permission from The Crop Science Society of America.). (B) Grain yield at 12 locations in SE Australia in 2003 of the low-$\Delta^{13}C$ cultivar Drysdale relative to the grain yield of a widely-grown variety currently recommended in that region (check variety). (Data in Fig. 6B courtesy of Agritech Research Services) Solid lines indicate statistically significant relationships (P <0.05).

lowest-yielding environments and *c*. 3% in the highest-yielding environments. At the sites in Western Australia, with Mediterranean-type rainfall patterns, high-*A/T* lines had a small (*c*. 3%) yield advantage over low-*A/T* lines. Yield levels at these sites were restricted to a small range, 1.9–2.6 t ha^{-1}. High-yielding low-$\Delta^{13}C$ breeding lines that also combined good quality and disease resistance were submitted for more comprehensive evaluation, resulting in the recent release of two of these lines as commercial cultivars in eastern Australia: 'Drysdale' (in October 2002) and 'Rees' (in October 2003). The yield in 2003 in SE-Australia of Drysdale, relative to that of a current commercial cultivar recommended for the same region, is shown in Fig. 6B. Drysdale had a clear yield advantage in the lower range of yields. These low yield levels (*c*. 2–3 t ha^{-1}) are typical for this region.

In breeding Drysdale and Rees, backcrossing was used to correct the 'deficiency' of low *A/T* while retaining the many favourable traits assembled in the widely-grown recurrent parent. These traits included a host of quality attributes and high levels of protection against a broad spectrum of biotic stresses. Backcrossing is easiest with a highly-heritable, simple trait (Simmonds, 1979), but even though $\Delta^{13}C$ is undoubtedly a poly-genic trait, it has a very high heritability if some care is taken with sampling protocols (Condon *et al*., 1993; Rebetzke *et al*., 2002).

It is interesting that in the backcross population from which Drysdale and Rees were identified, selecting for high *A/T* carried no yield penalty in the Mediterranean-type environments of Western Australia nor at sites with higher yield levels of *c*. 6 t ha^{-1} in eastern Australia. This appears contrary to the results of many of the studies reviewed earlier. The reason for this is not clear, but results indicate that in this case low $\Delta^{13}C$ was not associated with slower growth (AG Condon and GJ Rebetzke, unpublished data). This may partly be explained by the fact that the average difference in leaf $\Delta^{13}C$ between the two sets of backcross lines was relatively small, only *c*. 0.8‰, but still large enough, in theory, to generate 15% greater *A/T* in low-$\Delta^{13}C$ genotypes. It may be that the difference in $\Delta^{13}C$ arose from two components: one component resulting from a small reduction in stomatal conductance, having very little effect on *A* (Fig. 1), and a second component resulting from a small increase in photosynthetic capacity that did not penalize the rate of crop growth but countered any potential effect on *A* of the reduction in stomatal conductance. The net outcome of such a scenario would be a gain in *A/T* from both components and very little, if any, effect on crop growth rate in the absence of soil water deficit.

Advantages of measuring Δ^{13}C in leaves of well-watered plants

In the backcrossing programme just described, genotypic variation in $\Delta^{13}C$ was assessed using leaf material sampled early in crop growth from well-watered plants. This may appear paradoxical, given that the objective was to increase grain yield in environments where water limitation tends to increase as the growing season progresses. Why wasn't screening for variation in $\Delta^{13}C$ done during post-anthesis stress using the grain, for instance? There were two 'practical' reasons. One is that $\Delta^{13}C$ measured early in the season has been shown to provide the highest repeatability and heritability (Condon and Richards, 1992). Nearer to anthesis, substantial differences in the extent of soil water depletion may develop among genotypes (Condon and Richards, 1993; Condon *et al*., 1993) and therefore substantial differences in stress-induced stomatal closure that lead to genotype×environment ($G\times E$) effects on $\Delta^{13}C$. High $G\times E$ reduces heritability. Another reason for using early-formed leaf material is that it allows the use as parents in backcrossing, plants sampled for $\Delta^{13}C$ just a few weeks earlier. This saves time going through to the

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

next generation, and speed is often vital in backcrossing programmes (Simmonds, 1979).

There is another important reason for using unstressed leaf material. For cereals, processes in the weeks before flowering are often much more important for yield than processes during grain filling (Passioura, 1977; Fischer, 1981). This is because variation in grain yield is more strongly related to variation in grain number than grain size. Grain number is effectively set in the period leading up to anthesis. In Australia and many other rain-fed cereal cropping regions, it is common for soil-water depletion to start during the phase before anthesis and for substantial soil-water deficits to have developed by anthesis that may limit grain number. Sampling for variation in early-leaf $\Delta^{13}C$ provides an estimate of variation in *A/T* at the start of the phase of rapid crop growth and water use leading up to anthesis. Variation in *A/T* will be most influential during the early part of this phase because this is when plants are relatively unstressed, the stomata of all genotypes are more open and a large proportion of final C is assimilated by the crop.

Advantages of measuring $\Delta^{13}C$ of grain

For cereals such as wheat or barley grown in Mediterranean-type environments, higher yields have often been associated with high $\Delta^{13}C$, even in relatively dry locations (Voltas *et al*., 1999; Merah *et al*., 2001; Araus *et al*., 2003). The $\Delta^{13}C$ of grain rather than of leaves was measured in many of these studies. This association between grain $\Delta^{13}C$ and yield may well prove useful in selecting for higher yield in Mediterranean-type environments (Royo *et al*., 2002; Araus *et al*., 2003), but it is not clear how to interpret these relationships. Grain $\Delta^{13}C$ is a much more ambiguous measurement than $\Delta^{13}C$ measured on unstressed leaves.

It is likely that the association between high grain $\Delta^{13}C$ and high yield in Mediterranean environments is due to a combination of several factors. If high grain $\Delta^{13}C$ values mirror high values of $\Delta^{13}C$ at early stages of growth, then high grain $\Delta^{13}C$ and high yield may reflect faster growth rate throughout crop development (Condon *et al*., 1993). But often the correlation between grain and leaf $\Delta^{13}C$ values has been found to be relatively weak (Condon and Richards, 1992; Merah *et al*., 2001), so other factors may also be important. It may be that high grain $\Delta^{13}C$ values reflect greater reliance for grain filling on pre-anthesis stem reserves laid down earlier in the season when plants were less stressed. It is likely that, to fill their grain, genotypes that have used more of the soil water store at anthesis will be more reliant on reserves laid down before anthesis, when $\Delta^{13}C$ values were relatively high (Condon and Hall, 1997; Condon *et al*., 2002). Within an environment, high grain $\Delta^{13}C$ may indicate greater access to soil moisture during grain filling because of more extensive rooting (White *et al*., 1990; Condon *et al*., 1993), or because of earlier flowering, or it may reflect the ability to maintain stomata more open after anthesis despite increasing soil and atmospheric water stress (Condon *et al*., 1993). Whatever the cause of high grain $\Delta^{13}C$, any of these characteristics (fast crop growth rate, ability to remobilize stored reserves, earlier flowering, better water extraction, stomatal insensitivity to water deficit) are likely to be useful for cereals in Mediterranean-type environments.

Opportunities to improve yield by breeding for high A/T in other species

Exploiting high *A/T* in breeding for greater agronomic water-use efficiency is complicated for cereals by an association between high *A/T* and slow crop growth rate. This seems not to be the case for groundnut (*Arachis hypogaea* L.). For this species, field studies in both well-watered and water-limited environments consistently show greater biomass production to be associated with higher *A/T* (Wright *et al*., 1993; Nautiyal *et al*., 2002). In groundnut, variation in photosynthetic capacity accounts for a large proportion of the variation in *A/T* (Nageswara Rao *et al*., 1995). Importantly, high photosynthetic capacity in this species does not appear to be associated with a slower rate of leaf area growth. Effective selection for high *A/T* in groundnut is achieved via selection for low specific leaf area (*SLA*), a less expensive alternative to $\Delta^{13}C$. However, one complication still remains, which is a tendency for low *SLA* (high *A/T*) to be associated with low *HI* (Wright *et al*., 1993). The problem is being overcome by applying concurrent selection pressure for low *SLA* and high *HI* (Nigam *et al*., 2001).

There may be species in addition to groundnut in which high photosynthetic capacity is associated with faster crop growth and higher *A/T*. Likely candidates may be other grain legumes because they have the capacity for symbiotic nitrogen fixation (Condon and Hall, 1997). Limited screening of cowpea (*Vigna unguiculata* (L.) Walp.) (Ashok *et al*., 1999) and common bean (*Phaseolus vulgaris* L.) (Ehleringer *et al*., 1991) germplasm indicates that variation in stomatal conductance may be the dominant source of variation in *A/T* for these species. More extensive screening is warranted in these and other legumes.

Identifying species or genotypes of species in which high photosynthetic capacity is associated with faster crop growth and higher *A/T* could be done by combining measurements of $\Delta^{13}C$ with measurements of conductance and/or photosynthetic capacity. There are techniques available for detecting genotypic variation in conductance directly and rapidly using viscous-flow porometers (Rebetzke *et al*., 2001) or indirectly using oxygen isotope composition of dry matter (Farquhar *et al*., 1997) or canopy temperature (Fischer *et al*., 1998). Measurements of specific leaf area (Nageswara Rao *et al*., 1995) or leaf chlorophyll concentration (Rao *et al*., 2001) may be effective means of characterizing variation in photosynthetic capacity. Of course there may also be cropping environments for some crop species

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

where the slow crop growth rate and high *A/T* associated with low stomatal conductance is desirable because soil water is conserved for the critical flowering phase. The use of crop simulation models may be useful in identifying such circumstances.

For sunflower (*Helianthus annuus* L.), early research indicated that photosynthetic capacity was the major source of variation in *A/T* (Virgona and Farquhar, 1996), but more recent evidence indicates large genotypic variation in stomatal conductance as well (Lambrides *et al*., 2004). The same research group has also been evaluating the effect on hybrid sunflower yield of breeding and selection for low $\Delta^{13}C$. Progeny contrasting for $\Delta^{13}C$ were selected from a cross between high-$\Delta^{13}C$ and low-$\Delta^{13}C$ parents. These high-$\Delta^{13}C$ and low-$\Delta^{13}C$ progeny were then crossed with the same male tester lines and the resulting low-$\Delta^{13}C$ and high-$\Delta^{13}C$ pools of hybrids were evaluated for $\Delta^{13}C$ and yield in rain-fed trials in southern Queensland, Australia. The low-$\Delta^{13}C$ pool of hybrids significantly out-yielded the high-$\Delta^{13}C$ pool of hybrids in three of the four environments. The average yield advantage over all four environments was 25% (CJ Lambrides, SC Chapman, R Shorter, unpublished data).

Analysing large numbers of breeding lines for variation in $\Delta^{13}C$ is an expensive exercise. These costs could be reduced by making use of DNA-assisted screening techniques. There have been some studies that have identified potential QTLs for $\Delta^{13}C$ in various species (Specht *et al*., 2001; Thumma *et al*., 2001; Teulat *et al*., 2002). The majority of these studies has been done on plants subjected to soil water deficit, so it is difficult to determine whether the QTLs are associated with 'constitutive' or 'inducible' variation in $\Delta^{13}C$. Variation in $\Delta^{13}C$ may be caused by variation in stomatal conductance or photosynthetic capacity. QTL analysis might also be usefully applied specifically for these traits, since variation in conductance and capacity may have different impacts on crop growth and yield, depending on the species. QTLs associated with variation in stomatal conductance have been identified in species such as rice (Price *et al*., 1997; Ishimaru *et al*., 2001).

Greater early vigour to improve agronomic water-use efficiency

To this point, this review has focused on improving agronomic water-use efficiency by manipulating leaf-level water-use efficiency, primarily through the use of $\Delta^{13}C$. However, there are other strategies that also offer promise (reviewed by Richards *et al*., 2002). As discussed earlier, an important strategy to improve agronomic water-use efficiency in cropping environments where the soil surface is frequently rewetted is to restrict water lost by evaporation from the soil surface. This maximizes crop transpiration and improves the ratio *T/ET* (Condon and Richards, 1993; Richards *et al*., 2002). For cereals, indeed any crop, a reduction in soil evaporation is most easily achieved through the rapid development of leaf area to shade the soil surface from direct solar radiation. Good stand establishment and vigorous early plant growth will both contribute to rapid leaf area development.

Good stand establishment is best achieved by plants that reach the soil surface quicker if seed is sown relatively shallow, and that reach the soil surface much more consistently if seed is sown deep, such as when farmers are seeding into moisture below a dry topsoil. In wheat, the widely-used GA-insensitive dwarfing genes *Rht-B1b* (*Rht1*) and *Rht-D1b* (*Rht2*) have had a major impact on agronomic water-use efficiency by improving *HI* and crop standability. The latter feature has been most important in irrigated cropping systems, but the same genes strongly inhibit the expression of long coleoptiles that may be an important attribute for rain-fed systems (Ellis *et al*., 2004). Alternative, GA-sensitive dwarfing genes exist in wheat that allow the expression of much longer coleoptiles (from tall wheats) in plants with semi-dwarf stature and high *HI* (Rebetzke *et al*., 1999).

Early vigour (fast leaf area development) is an important adaptation of barley and durum wheat to terminal drought in Mediterranean environments (Van Oosterum and Acevedo, 1992; Lopez-Castaneda and Richards, 1994; Villegas *et al*., 2000) because it improves the ratio *T/ET* and encourages growth when evaporative demand is low, giving higher *A/T*. Traits important for vigorous early growth in cereals were identified by comparing barley with bread wheat (Lopez-Castaneda *et al*., 1995). This comparison showed that large embryo size, high *SLA*, and growth of a large coleoptile tiller were important attributes of barley, which is characterized by very high early vigour, but these traits were lacking in semi-dwarf wheats. Extensive screening of a collection of tall wheats revealed excellent sources of each of these traits (Richards and Lukacs, 2002). In a targeted breeding programme, these were combined to produce a new parental line with early leaf area growth double that of current Australian semi-dwarf varieties (Richards *et al*., 2002). High-vigour backcross lines with this parent as the vigour donor out-yielded low-vigour lines from the same population by up to 13% in favourable Mediterranean-type environments (*c*. 280–450 mm in-season rainfall), but there was no difference in yield in drier environments with less than 250 mm in-season rainfall (Table 1; Botwright *et al*., 2002). The sets of vigour backcross lines grown by Botwright *et al*. (2002) were still much less vigorous than the tall vigour donor, probably because the recurrent parent contained a GA-insensitive dwarfing gene. The high vigour traits of large embryo, large coleoptile tillers, and greater *SLA* are now being combined in GA-sensitive semi-dwarf backgrounds so as to achieve early vigour much more like barley (Richards *et al*., 2002). Field-based screening for early leaf area growth is subject to large genotype×environment interaction and hence poor heritability (Annicchiarico and

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

Table 1. *Growth and yield of two sets of BC_2:F_5-derived wheat breeding lines contrasting for early vigour and sown at two locations in Western Australia in 1999 (from Botwright* et al.*, 2002)*

Location	Vigour class	Leaf area index 50 DAS[a]	Anthesis biomass (g m^{-2})	Final biomass (g m^{-2})	Grain yield (g m^{-2})
Wongan Hills[b]	High vigour	0.37	453	678	337
	Low vigour	0.32*[c]	405*	573**	293***
Merredin[b]	High vigour	0.39	511	634	266
	Low vigour	0.30**	527n.s.	574**	247**

[a] DAS, days after sowing.
[b] May-November rainfall 453 mm at Wongan Hills, 274 mm at Merredin.
[c] *P <0.05; ** P <0.01; *** P <0.001; n.s., not significant (for contrasts within locations).

Pecetti, 1998; Rebetzke and Richards, 1999). To counter this, Richards *et al*. (2002) advocated selection using plants grown in more repeatable conditions and the use of leaf width as an indirect selection tool for embryo size. This is because of the high heritability of leaf width and its strong association with early leaf area.

Combining greater early vigour and higher A/T

The outcomes of the simulation study summarized earlier (Fig. 5) indicated that combining greater early vigour and higher *A/T* may give substantially greater average yield gains in many rain-fed environments, but especially in environments with large temporal variability in rainfall. Is such a combination of traits possible? High early vigour in cereals appears to be partly associated with high *SLA* (Lopez-Castaneda *et al*., 1995; Rebetzke *et al*., 2004). To the extent that high *SLA* is reflected in low photosynthetic capacity (via less N per unit leaf area), then high *SLA* may also be reflected in higher $\Delta^{13}C$. This suggests that if high vigour and low-$\Delta^{13}C$ are to be successfully combined, then a tendency to higher *SLA* will need to be avoided during selection. This may be desirable for other reasons. *SLA* has relatively low heritability in cereals (Rebetzke *et al*., 2004), so its value as a selection tool for high early vigour may be limited. Also, simulation modelling of the impact of selecting for higher *SLA* in wheat indicates that maintaining N supply may be an important prerequisite for sustaining yield gains achieved via this trait (Asseng *et al*., 2003). The associations between early vigour in cereals and traits other than high *SLA*, such as embryo size and coleoptile tiller appearance (Liang and Richards, 1994), indicate that there may be other ways to achieve higher early vigour that could allow co-selection for lower values of $\Delta^{13}C$.

Conclusion

There is now growing evidence that targeting specific traits in a breeding programme may lead to higher agronomic water-use efficiency. It is also clear that the effects of any one trait must be considered in the context of the environment in which the crop is to be grown. A particular trait, such as high *A/T*, may be associated with higher yield in one type of environment but may have no effect or even be detrimental in other environments. Breeding for high *A/T* using low-$\Delta^{13}C$ measured in the leaves of unstressed wheat plants has resulted in the release of new, higher-yielding varieties for eastern Australia. The new varieties result from a backcrossing programme targeting environments where stored soil moisture needs to be metered out from relatively early in the cropping season so as to maximize seed set and sustain seed growth. By contrast, for cereals growing in Mediterranean-type cropping regions and irrigated environments, higher yield appears to be associated with high-$\Delta^{13}C$ of grain. For Mediterranean-type regions this association may, in part, be a result of an association between high-$\Delta^{13}C$ and fast crop growth rate. Fast crop growth rate is reflected in the vigorous development of leaf area to shade the soil surface, a key trait for cropping environments where the soil surface is frequently rewetted and large gains in *T/ET* can be made. Effective selection protocols for faster leaf area growth have been devised and shown to be successful for yield improvement in Mediterranean-type environments. It is possible that even greater yield gains may be achievable in these and other rain-fed environments if fast crop growth rate can be combined with high *A/T*. Breeding has been initiated to combine these two traits in bread and durum wheat, but this may prove difficult if an association between high vigour and high-$\Delta^{13}C$ cannot be broken.

A strategy that sets out deliberately to target a combination of high early vigour and high *A/T* in wheat may not be necessary in other species. It will depend strongly on the extent and timing of any water limitation in relation to developmental phases critical for yield determination, and whether there is an association between high *A/T* and slow crop growth rate. Such an association is likely to depend on whether variation in *A/T* is due to variation in stomatal conductance or photosynthetic capacity and the effects of these two components on crop growth rate. Relatively simple techniques are now available for characterizing variation in stomatal conductance and photosynthetic capacity. Other interactions may also come into play. In groundnut, *A/T* is positively associated with crop biomass production, but negatively associated with *HI*. Breeding progress is being made in this species by applying concurrent selection pressure for high *A/T* and high *HI*. Progress may be more rapid if the reasons for the association between *A/T* and *HI* were better understood.

Acknowledgements

The authors acknowledge the Grains Research and Development Corporation (GRDC) for providing funds towards the research on

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

wheat in Australia. Agritech Research Services and GRDC are acknowledged for permission to publish data in Fig. 6B.

References

Annicchiarico P, Pecetti L. 1998. Yield versus morphophysiological trait-based criteria for selection of durum wheat in a semi-arid Mediterranean region (northern Syria). *Field Crops Research* **59,** 163–173.

Araus JL, Villegas D, Aparicio N, Garcia del Moral LF, El Hani S, Rharrabti Y, Ferrio JP, Royo C. 2003. Environmental factors determining carbon isotope discrimination and yield in durum wheat under Mediterranean conditions. *Crop Science* **43,** 170–180.

Ashok, Hussain ISA, Prasad TG, Kumar MU, Rao RCN, Wright GC. 1999. Variation in transpiration efficiency and carbon isotope discrimination in cowpea. *Australian Journal of Plant Physiology* **26,** 503–510.

Asseng S, Turner NC, Botwright TL, Condon AG. 2003. Evaluating the impact of a trait for increased specific leaf area on wheat yields using a crop simulation model. *Agronomy Journal* **95,** 10–19.

Botwright TL, Condon AG, Rebetzke GJ, Richards RA. 2002. Field evaluation of early vigour for genetic improvement of grain yield in wheat. *Australian Journal of Agricultural Research* **53,** 1137–1145.

Brugnoli E, Farquhar GD. 2000. Photosynthetic fractionation of carbon isotopes. In: Leegood RC, Sharkey TD, von Caemmerer S, eds. *Photosynthesis: physiology and metabolism*. Advances in Photosynthesis, Vol. 9. Dordrecht: Kluwer Academic Publishers, 399–434.

Condon AG, Farquhar GD, Richards RA. 1990. Genotypic variation in carbon isotope discrimination and transpiration efficiency in wheat. Leaf gas exchange and whole plant studies. *Australian Journal of Plant Physiology* **17,** 9–22.

Condon AG, Hall AE. 1997. Adaptation to diverse environments: genotypic variation in water-use efficiency within crop species. In: Jackson LE, ed. *Agricultural ecology*. San Diego, CA: Academic Press, 79–116.

Condon AG, Richards RA. 1992. Broad sense heritability and genotype×environment interaction for carbon isotope discrimination in field-grown wheat. *Australian Journal of Agricultural Research* **43,** 921–934.

Condon AG, Richards RA. 1993. Exploiting genetic variation in transpiration efficiency in wheat: an agronomic view. In: Ehleringer JR, Hall AE, Farquhar GD, eds. *Stable isotopes and plant carbon-water relations*. San Diego, CA: Academic Press, 435–450.

Condon AG, Richards RA, Farquhar GD. 1987. Carbon isotope discrimination is positively correlated with grain yield and dry matter production in field-grown wheat. *Crop Science* **27,** 96–1001.

Condon AG, Richards RA, Farquhar GD. 1993. Relationships between carbon isotope discrimination, water-use efficiency and transpiration efficiency for dryland wheat. *Australian Journal of Agricultural Research* **4,** 1693–1711.

Condon AG, Richards RA, Rebetzke GJ, Farquhar GD. 2002. Improving intrinsic water-use efficiency and crop yield. *Crop Science* **42,** 122–131.

Cooper PJM, Gregory PJ, Keatinge JDH, Brown SC. 1987. Effects of fertilizer, variety and location on barley production under rainfed conditions in northern Syria. 2. Soil water dynamics and crop water use. *Field Crops Research* **16,** 67–84.

Ehleringer JR, Klassen S, Clayton C, Sherrill D, Fuller-Holbrook M, Fu QA, Cooper TA. 1991. Carbon isotope discrimination and transpiration efficiency in common bean. *Crop Science* **31,** 1611–1615.

Ellis MH, Rebetzke GJ, Chandler P, Bonnett D, Spielmeyer W, Richards RA. 2004. The effect of different height reducing genes on the early growth characteristics of wheat. *Functional Plant Biology* **31,** 583–589.

Farquhar GD, O'Leary MH, Berry JA. 1982. On the relationship between carbon isotope discrimination and the intercellular carbon dioxide concentration in leaves. *Australian Journal of Plant Physiology* **9,** 121–137.

Farquhar GD, Richards RA. 1984. Isotopic composition of plant carbon correlates with water-use efficiency of wheat genotypes. *Australian Journal of Plant Physiology* **11,** 539–552.

Farquhar GD, Ehleringer JR, Hubick KT. 1989. Carbon isotope discrimination and photosynthesis. *Annual Review of Plant Physiology and Plant Molecular Biology* **40,** 503–537.

Farquhar GD, Henry BK, Styles JM. 1997. A rapid on-line technique for determination of oxygen isotope composition of nitrogen-containing organic matter and water. *Rapid Communications in Mass Spectrometry* **11,** 1554–1560.

Fischer RA. 1981. Optimizing the use of water and nitrogen through breeding of crops. *Plant and Soil* **58,** 249–278.

Fischer RA, Rees D, Sayre KD, Lu Z-M, Condon AG, Larque Saavedra A. 1998. Wheat yield progress is associated with higher stomatal conductance, higher photosynthetic rate and cooler canopies. *Crop Science* **38,** 1467–1475.

Hall AE, Richards RA, Condon AG, Wright GC, Farquhar GD. 1994. Carbon isotope discrimination and plant breeding. In: Janick J, ed. *Plant breeding reviews*, Vol. 12. New York: Wiley, 81–113.

Hamdy A, Ragab R, Scarascia-Mugnozza E. 2003. Coping with water scarcity: water saving and increasing water productivity. *Irrigation and Drainage* **52,** 3–20.

Ishimaru K, Shirota K, Higa M, Kawamitsu Y. 2001. Identification of quantitative trait loci for adaxial and abaxial stomatal frequencies in *Oryza sativa*. *Plant Physiology and Biochemistry* **39,** 173–177.

Keatinge JDH, Cooper PJM. 1983. Kabuli chickpea as a winter-sown crop in northern Syria: moisture relations and crop productivity. *Journal of Agricultural Science (Cambridge)* **100,** 667–680.

Lambrides CJ, Chapman SC, Shorter R. 2004. Surveys of carbon isotope discrimination in sunflower reveal considerable genetic variation, a strong association with transpiration efficiency and evidence of cytoplasmic inheritance. *Crop Science* **44,** 1642–1653.

Liang YL, Richards RA. 1994. Coleoptile tiller development is associated with fast early vigour in wheat. *Euphytica* **80,** 119–124.

Long SP, Bernacchi CJ. 2003. Gas exchange measurements, what can they tell us about the underlying limitations to photosynthesis? Procedures and sources of error. *Journal of Experimental Botany* **54,** 2393–2401.

Lopez-Castaneda C, Richards RA. 1994. Variation in temperate cereals in rainfed environments. III. Water use and water-use efficiency. *Field Crops Research* **39,** 85–98.

Lopez-Castaneda C, Richards RA, Farquhar GD. 1995. Variation in early vigor between wheat and barley. *Crop Science* **35,** 472–479.

Merah O, Deleens E, Souyris I, Nachit M, Monneveux P. 2001. Stability of carbon isotope discrimination and grain yield in durum wheat. *Crop Science* **41,** 677–681.

Morgan JA, LeCain DR. 1991. Leaf gas exchange and related leaf traits among 15 winter wheat genotypes. *Crop Science* **31,** 443–448.

Nageswara Rao RC, Udayakumar M, Farquhar GD, Talwar HS, Prasad TG. 1995. Variation in carbon isotope discrimination and its relationship to specific leaf area and ribulose-1,5-bisphosphate carboxylase content in groundnut genotypes. *Australian Journal of Plant Physiology* **22,** 545–551.

Nautiyal PC, Nageswara Rao RC, Joshi YC. 2002. Moisture-deficit-induced changes in leaf-water content, leaf carbon exchange rate and biomass production in groundnut cultivars differing in specific leaf area. *Field Crops Research* **74,** 67–79.

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024

Nigam SN, Upadhyaya HD, Chandra S, Rao RCN, Wright GC, Reddy AGS. 2001. Gene effects for specific leaf area and harvest index for three crosses of groundnut (*Arachis hypogaea*). *Annals of Applied Biology* **139,** 301–306.

Passioura JB. 1977. Grain yield, harvest index and water use of wheat. *Journal of the Australian Institute of Agricultural Science* **43,** 117–120.

Price AH, Young EM, Tomos AD. 1997. Quantitative trait loci associated with stomatal conductance, leaf rolling and heading date mapped in upland rice (*Oryza sativa*). *New Phytologist* **137,** 83–91.

Rao RCN, Talwar HS, Wright GC. 2001. Rapid assessment of specific leaf area and leaf chlorophyll in peanut (*Arachis hypogaea* L.) using a chlorophyll meter. *Journal of Agronomy and Crop Science* **186,** 175–182.

Rebetzke GJ, Botwright TL, Moore CS, Richards, RA, Condon AG. 2004. Genotypic variation in specific leaf area for genetic improvement of early vigour in wheat. *Field Crops Research* **88,** 179–189.

Rebetzke GJ, Condon AG, Richards RA, Farquhar GD. 2002. Selection for reduced carbon-isotope discrimination increases aerial biomass and grain yield of rainfed bread wheat. *Crop Science* **42,** 739–745.

Rebetzke GJ, Condon AG, Richards RA, Read JJ. 2001. Phenotypic variation and sampling for leaf conductance in wheat (*Triticum aestivum* L.) breeding populations. *Euphytica* **121,** 335–341.

Rebetzke GJ, Richards RA. 1999. Genetic improvement of early vigour in wheat. *Australian Journal of Agricultural Research* **50,** 291–301.

Rebetzke GJ, Richards RA, Fischer VM, Mickelson BJ. 1999. Breeding long coleoptile, reduced height wheats. *Euphytica* **106,** 159–168.

Richards RA, Condon AG. 1993. Challenges ahead in using carbon isotope discrimination in plant-breeding programs. In: Ehleringer JR, Hall AE, Farquhar GD, eds. *Stable isotopes and plant carbon-water relations*. San Diego, CA: Academic Press, 451–462.

Richards RA, Lukacs Z. 2002. Seedling vigour in wheat—sources of variation for genetic and agronomic improvement. *Australian Journal of Agricultural Research* **53,** 41–50.

Richards RA, Rebetzke GJ, Condon AG, van Herwaarden AF. 2002. Breeding opportunities for increasing efficiency of water use and crop yield in temperate cereals. *Crop Science* **42,** 111–121.

Royo C, Villegas D, Garcia del Moral LF, Elhani S, Aparicio N, Rharrabti Y, Araus JL. 2002. Comparative performance of carbon isotope discrimination and canopy temperature depression as predictors of genotype differences in durum wheat yield in Spain. *Australian Journal of Agricultural Research* **53,** 561–569.

Sayre KD, Acevedo E, Austin RB. 1995. Carbon isotope discrimination and grain yield for three bread wheat germplasm groups grown at different levels of water stress. *Field Crops Research* **41,** 45–54.

Simmonds NW. 1979. *Principals of crop improvement*. London: Longman.

Specht JE, Chase K, Macrander M, Graef GL, Chung J, Markwell JP, Germann M, Orf JH, Lark KG. 2001. Soybean response to water: a QTL analysis of drought tolerance. *Crop Science* **41,** 493–509.

Stapper M, Harris HC. 1989. Assessing the productivity of wheat genotypes in a mediterranean climate, using a simulation model. *Field Crops Research* **20,** 129–152.

Tanner CB, Sinclair TR. 1983. Efficient water use in crop production: research or re-search? In: Taylor HM, Jordan WR, Sinclair TR, eds. *Limitations to efficient water use in crop production*. Madison, WI: ASA-CSSA-SSSA, 1–27.

Teulat B, Merah O, Sirault X, Bories C, Waugh R, This D. 2002. QTLs for grain carbon isotope discrimination in field-grown barley. *Theoretical and Applied Genetics* **106,** 118–126.

Thumma BA, Naidu BP, Chandra A, Cameron DF, Bahnisch LM, Liu C. 2001. Identification of causal relationships among traits related to drought resistance in *Stylosanthes scabra* using QTL analysis. *Journal of Experimental Botany* **52,** 203–214.

Van Oosterom EJ, Acevedo E. 1992. Adaptation of barley (*Hordeum vulgare* L.) to harsh Mediterranean environments. 1. Morphological traits. *Euphytica* **62,** 1–14.

Villegas D, Aparicio N, Nachit MM, Araus JL, Royo C. 2000. Photosynthetic and developmental traits associated with genotypic differences in durum wheat yield across the Mediterranean basin. *Australian Journal of Agricultural Research* **51,** 891–901.

Virgona JM, Farquhar GD. 1996. Genotypic variation in relative growth rate and carbon isotope discrimination in sunflower is related to photosynthetic capacity. *Australian Journal of Plant Physiology* **23,** 227–236.

Voltas J, Romagosa I, Lafarga A, Armesto AP, Sombrero A, Araus JL. 1999. Genotype by environment interaction for grain yield and carbon isotope discrimination of barley in Mediterranean Spain. *Australian Journal of Agricultural Research* **50,** 1263–1271.

Wang HX, Liu CM, Zhang L. 2002. Water-saving agriculture in China: an overview. *Advances in Agronomy* **75,** 135–171.

White JW, Castillo JA, Ehleringer J. 1990. Associations between productivity, root growth and carbon isotope discrimination in *Phaseolus vulgaris* under water deficit. *Australian Journal of Plant Physiology* **17,** 189–198.

Wright GC, Hubick KT, Farquhar GD, Nageswara Rao RC. 1993. Genetic and environmental variation in transpiration efficiency and its correlation with carbon isotope discrimination and specific leaf area in peanut. In: Ehleringer JR, Hall AE, Farquhar GJ, eds. *Stable isotopes and plant carbon-water relations*. San Diego, CA: Academic Press, 245–267.

Downloaded from https://academic.oup.com/jxb/article/55/407/2447/496068 by UAMS Library user on 14 March 2024